MTH2008 Real Analysis: Graded Formative Assessment 1

This Graded Formative Assessment Sheet covers the material presented in Weeks 1 and 2.

If you would like to receive feedback on your work, please hand in solutions to Questions 2 and 6 by 12:00 noon on Wednesday 07 October 2026. As it is optional to submit solutions to these two assigned questions for feedback, if your submission is late, it will be at your tutors discretion as to if it is graded or not.

Note yours solutions to the two assigned questions, in total, should not exceed 750 words, and that this is not a target. Indeed, if you find yourself writing more than this, take a step back and think about your solution and your approach to the problem.

Each of the two assigned questions will be graded on a 10-point scale and your mark on ELE will be scaled by a factor of 5 so as to be a mark out or 100.

Please also ensure you upload your solutions to ELE in a single PDF (Portable Document Format) unless otherwise instructed by your tutor. As it is optional to submit solutions to the two assigned questions for feedback, if your file is not uploaded in a single PDF, it will be at your tutors discretion as to if it is graded or not.

Note also, that Summative Assessment 1 will only consist of questions taken directly from this assessment.

Learning outcomes for Graded Formative Assessment 1

Axioms that define the real number, sets and set operations, the topology of the real line, functions of a single variable, and continuity and limits.

Assessment criteria for Graded Formative Assessment 1

If this were an assessment which counted towards your final grade, to achieve a pass mark, you will have needed to demonstrate that you know the basic definitions of the subject and can verify whether or not they hold in some familiar cases. To achieve full marks, you will have needed to demonstrate a thorough understanding of the concepts by relating them to more elaborate mathematical contexts. You will also have needed to demonstrate the ability to construct mathematical arguments and to communicate them effectively through clearly explained and logically correct proofs with accurate use of appropriate mathematical notation and terminology.

Statement on use of GenAI tools for Graded Formative Assessment 1

As Graded Formative Assessment 1 has the option for you to submit your solutions to a selection of questions (see above for precise details) in order for you to gain feedback, it is advised that you treat this assessment with the classification of being AI-Minimal. This means that you may use AI tools for checking spelling and grammar mistakes only, with no other impact on the structure or content of the assessment. This is because using AI tools outside of these uses prevents fair assessment of your ability to achieve the module learning outcomes. As such, when writing your assessment, try not to use AI tools in the following ways:

(i) for uses other than checking your spelling and grammar,

(ii) to translate more than a word or short phrase into English,

(iii) to upload sensitive or identifying material to an AI tool, and

(iv) to present material that has been generated by AI as your own work or the work of someone else.

1. Let Y be a set of positive real numbers with supremum $\alpha = \sup Y \in \mathbb{R}$ and let $Z = \{x^2 - \sqrt{5} \,|\, x \in Y\}$. Justifying your answer, determine the supremum of Z.

Solution: We claim that $\sup Z = \alpha^2 - \sqrt{5}$. Since $\alpha = \sup Y$, then $x \leq \alpha$, which exists by the completeness axiom, for every $x \in Y$, hence $x^2 - \sqrt{5} \leq \alpha^2 - \sqrt{5}$. Thus, since Y is non-empty, Z is non-empty, and since our preceding argumnets give us that Z is bounded from above, Z has a supremum. Now suppose that $\sup Z = \beta < \alpha^2 - \sqrt{5}$, then $x^2 - \sqrt{5} \leq \beta$ for all $x \in Y$, or in other words,

$$x \leq \sqrt{\beta + \sqrt{5}},$$

for all $x \in Y$. However, since $\beta < \alpha^2 - \sqrt{5}$, and since X is a set of positive real numbers, meaning that α is a non-negative real number, we have that

$$x \leq \sqrt{\beta + \sqrt{5}} < \sqrt{\alpha^2 - \sqrt{5} + \sqrt{5}} = \alpha.$$

for all $x \in Y$, contradicting the fact that $\alpha = \sup Y$.

2. Let S be a bounded non-empty subset of $\mathbb{R}$, and let $T = \{-s \mid s \in S\}$. Prove that $-\inf S = \sup T$.

Solution: Let $\alpha = \inf S$, which exists by the completeness axiom, then $s \geq \alpha$ for all $s \in S$, and if $a > \alpha$, then there exists $s_0 \in S$ such that $s_0 < a$. This implies that, $-s \leq -\alpha$ for all $s \in S$ and if $-a < -\alpha$, then there exists $s_0 \in S$ with $-s_0 > -a$. In other words, $-\alpha = \sup T$.

3. Let S and T be nonempty sets of real numbers and define

$$S - T = \{s - t \mid s \in S \text{ and } t \in T\}.$$

Prove that

$$\sup(S - T) \leq \sup S - \inf T$$

if S and T are bounded.

Solution: Assume S and T are bounded, and set

$$M = \sup S \quad \text{and} \quad m = \inf T.$$

Observe that these exist by the completeness axiom, and note that since S and T are non-empty, there exists $s \in S$ and $t \in T$, and so $S - T$ is non-empty since $s - t \in S - T$. Now, if $x \in S - T$, then $x = s - t$ for some $s \in S$ and $t \in T$. Since $s \leq M$ and $m \leq t$, we have that $x = s - t \leq M - m$. Hence $M - m$ is an upper bound of $S - T$, and so $\sup(S - T)$ exists and $\sup(S - T) \leq M - m$. Therefore, $\sup(S - T) \leq \sup S - \inf T$, as required.

4. Let S and T be nonempty bounded sets of real numbers and define

$$S - T = \{s - t \mid s \in S \text{ and } t \in T\}.$$

Prove, or give a counterexample to the equality

$$\sup(S - T) = \sup S - \inf T.$$

Solution: By the previous question, we know that $\sup(S - T) \leq \sup S - \inf T$. To prove that this is in fact an equality, let ϵ denote a positive real number. Since $M = \sup S$, there exists $s \in S$ such that $s > M - \epsilon/2$. Similarly, since $m = \inf T$, there exists $t \in T$ such that $t < m + \epsilon/2$. Therefore,

$$s - t > \left(M - \frac{\epsilon}{2}\right) - \left(m + \frac{\epsilon}{2}\right) = M - m - \epsilon.$$

In other words, $S - T$ contains an element exceeding $M - m - \epsilon$. Since this holds for every $\epsilon > 0$, we have that $M - m$ is the least upper bound of $S - T$. Hence

$$\sup(S - T) = M - m = \sup S - \inf T.$$

5. In $\mathbb{R}$, prove that the union of finitely many closed sets is closed. For this question, if needed, you may use, without proof, that finite intersections of open sets are open, and results such as De Morgan's law, and standard set theoretic relations.

Solution: Let $V_1, \ldots, V_n$ be closed sets and consider the set

$$V = \bigcup_{i=1}^{n} V_i.$$

Noting, by De Morgan's law, that

$$V^c = \bigcap_{i=1}^{n} V_i^c$$

and that every V_i^c is open (since the complement of a closed set is open), since finite intersections of open sets are open, V^c is open. Therefore, since the complement of an open set is closed, and the complement of the complement of a set is the set itself, we have that V is closed.

6. Justifying your answer, give an example showing that the union of infinitely many closed sets may fail to be closed.

Solution: For n a natural number, let V_n denote the closed interval $[1/n, 1-1/n]$, and observe that $\bigcup_{n\in\mathbb{N}} V_n = \bigcup_{n\in\mathbb{N}} [1/n, 1-1/n] = (0,1)$, which is open interval and hence not closed, as the only open and closed subsets of $\mathbb{R}$, are $\mathbb{R}$ and the empty set.

7. Prove that a set $S \subset \mathbb{R}$ is closed if and only if $S = \overline{S}$.

Solution: Assume that S is closed. This implies that S^c is open. Therefore, S^c is a neighbourhood of every point in S^c. This implies that no point of S^c can be a boundary point of S, since $S^c \cap S = \emptyset$. Therefore, S contains all of its boundary points, that is $\partial S \subset S$ and hence $\overline{S} \subset S$. Moreover, by definition, $S \subset \overline{S}$ and hence $S = \overline{S}$.

Now suppose that $\overline{S} = S$. In which case, since $\overline{S} = S \cup \partial S$, we have $\partial S \subseteq S$. Hence, no point $x_0 \in S^c$ is a boundary point. This implies that every point in S^c has a neighbourhood $U \subseteq S^c$, and therefore is an interior point of S^c. In other words, S^c is open and thus $S = (S^c)^c$ is closed.

8. Find an open covering $\mathcal{H}$ of the set

$$S = \left\{ \frac{(-1)^{n+1}}{n+1} \mid n \in \mathbb{N} \right\}$$

that cannot have a finite subcovering. Justify your answer.

Solution: For $k \in \mathbb{N}$ let

$$I_k = \left(-\frac{1}{2k}, -\frac{1}{2k+2} \right) \quad \text{and} \quad J_k = \left(\frac{1}{2k+1}, \frac{1}{2k-1} \right).$$

The following open covering would be a valid example,

$$\mathcal{H} = \{ I_k \big| k \in \mathbb{N} \} \cup \{ J_k \big| k \in \mathbb{N} \}.$$

Indeed, each interval in $\mathcal{H}$ covers one point only of S, which no other interval does, namely for $k \in \mathbb{N}$, the interval I_k covers the point $-1/(2k+1)$, and the interval J_k covers the point $1/(2k)$.

To see that this cover can not be reduced to a finite subcover, observe that for any finite subcollection from $\mathcal{H}$ we may let k be the largest natural number such that I_k or J_k is in the subcollection. In which case the points $-1/(2k+3)$ and $1/(2k+2)$ are not covered by the subcollection.

9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$, by

$$f(x) = x^2 - 2x + 1.$$

Using the ϵ-δ definition of a limit, prove that

$$\lim_{x \to 1} f(x) = 0.$$

Solution: Given $\epsilon > 0$, letting $\delta \in (0, \sqrt{\epsilon})$, for all $x \in \mathbb{R}$ with $|x-1| < \delta$,

$$|f(x) - 0| = |x^2 - 2x + 1| = |(x-1)^2| = |x-1|^2 < \delta^2 < \epsilon.$$

Thus, $\lim_{x\to 1} f(x) = 0$.

10. Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \mathbb{R}$. Prove that if there exists a real number $\epsilon > 0$ such that every open deleted δ-neighbourhood of x_0 contains points x_1 and x_2 with $|f(x_1) - f(x_2)| > \epsilon$, then f does not approach a finite limit as x approaches x_0.

Solution: We proceed by contradiction. Suppose that there exists $L \in \mathbb{R}$ such that

$$\lim_{x \to x_0} f(x) = L.$$

In which case, letting ϵ be as in the hypothesis of the question, there exists $\delta > 0$ such that $|f(x) - L| < \epsilon/2$ for all $x \in \mathbb{R}$ with $0 < |x - x_0| < \delta$. Further, by the hypothesis of the question, there exist x_1 and x_2 in $(x_0 - \delta, x_0 + \delta) - \{x_0\}$, such that

$$\epsilon < |f(x_1) - f(x_2)|.$$

This in tandem with the triangle inequality, implies that

$$\begin{aligned} \epsilon < |f(x_1) - f(x_2)| &= |f(x_1) - L + L - f(x_2)| \\ &\leq |f(x_1) - L| + |L - f(x_2)| < \epsilon/2 + \epsilon/2 < \epsilon, \end{aligned}$$

yielding a contradiction.

11. Suppose that a function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x_1 + x_2) = f(x_1) + f(x_2)$ for all $x_1, x_2 \in \mathbb{R}$ and that f is continuous at zero. Prove that f is continuous. You may use, without proof, standard results established in lectures, including the algebra of limits and the evaluation of standard limits.

Solution: By the hypothesis of the question, $f(0) = f(0 + 0) = f(0) + f(0)$, and thus $f(0) = 0$. Since f is continuous at 0, we have that

$$\lim_{h \to 0} f(h) = f(0) = 0.$$

Hence, for an arbitrary $x_0 \in \mathbb{R}$,

$$\lim_{x \to x_0} f(x) = \lim_{h \to 0} f(x_0 + h) = \lim_{h \to 0} (f(x_0) + f(h)) = \lim_{h \to 0} f(x_0) + \lim_{h \to 0} f(h) = f(x_0) + 0 = f(x_0).$$

Therefore, f is continuous at x_0. Since x_0 was chosen arbitrarily, f is continuous.

12. Let $f, g : (0, \infty) \to \mathbb{R}$ be defined, for $x \in (0, \infty)$, by

$$f(x) = \frac{x^2 + 2x + 3}{2x^3 + 1} \qquad \text{and} \qquad g(x) = \frac{1 - \sqrt{1/x}}{1 + \sqrt{1/x}}.$$

Using values of known limits and the algebra of limits, compute the value of the following limits.

(i) $\lim_{x \to 2} f(x)$

(ii) $\lim_{x \to 0^+} g(x)$

Solution:

(i) Since, for a constant $c \in \mathbb{R}$, we have that

$$\lim_{x \to 2} c = c \qquad \text{and} \qquad \lim_{x \to 2} cx = 2c,$$

by the algebra of limits, we have that

$$\lim_{x \to 2} x^2 + 2x + 3 = \left(\lim_{x \to 2} x \right)^2 + \left(\lim_{x \to 2} 2x \right) + \left(\lim_{x \to 2} 3 \right) = 2^2 + 2 \cdot 2 + 3 = 11$$

and

$$\lim_{x \to 2} 2x^3 + 1 = \left(\lim_{x \to 2} 2x \right) \left(\lim_{x \to 2} x \right)^2 + \left(\lim_{x \to 2} 1 \right) = 2 \cdot 2 \cdot 2^2 + 1 = 17 \neq 0.$$

Thus, by a second application of the algebra of limits,

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} \frac{x^2 + 2x + 3}{2x^3 + 1} = \frac{11}{17}.$$

(ii) Observing that, for all positive $x \in \mathbb{R}$

$$\frac{1-\sqrt{1/x}}{1+\sqrt{1/x}} = \frac{1-\sqrt{1/x}}{1+\sqrt{1/x}} \cdot \frac{1/\sqrt{1/x}}{1/\sqrt{1/x}} = \frac{1/\sqrt{1/x}-1}{1/\sqrt{1/x}+1} = \frac{\sqrt{x}-1}{\sqrt{x}+1}$$

Since,

$$\lim_{x\to 0^+} \sqrt{x} = 0,$$

by the algebra of limits, we conclude that

$$\lim_{x\to 0^+} g(x) = \lim_{x\to 0^+} \frac{1-\sqrt{1/x}}{1+\sqrt{1/x}} = \lim_{x\to 0^+} \frac{\sqrt{x}-1}{\sqrt{x}+1} = \frac{0-1}{0+1} = -1.$$